

Notes of Learning Theory

Yue Wu

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Outline

Intro - Learning Theory

Goal: Function Learning

Measurements: Excess Risk

Method: Bounding the Generalization Gap

Tools: Concentration and Suprema

Binary Classification

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Intro - Learning Theory

Recall classic statistics: parameter estimation (e.g. estimate the mean of a certain distribution)

- ▶ data: $\{x_i\}_{i=1}^n$
- ▶ hypothesis: distribution model of X
- ▶ method: statistics (e.g. $\hat{\theta} = \bar{X}$)
- ▶ analyze/evaluate: properties of the statistics ($\mathbb{E}(\hat{\theta} - \theta)$, ...) with $n/?$

Now in supervised learning: function estimation, i.e. parameter estimation under fixed functional form (e.g.)

- ▶ data: $\{(x_i, y_i)\}_{i=1}^n$
- ▶ hypothesis: distribution model of (X, Y) , $f : \mathcal{X} \rightarrow \mathcal{Y}$, $f \in \mathcal{F}$
- ▶ method: empirical risk minimization or ...

$$\hat{\mathcal{R}}(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i), \quad \hat{f} = \arg \min_{f \in \mathcal{F}} \hat{\mathcal{R}}(f)$$

- ▶ analyze/evaluate: excess risk with n, m, T

$$\mathcal{R}(\hat{f}) - \mathcal{R}(f^*).$$

More on Learning theory, see xx

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Measurements of a function estimation

How well model f approximates pair r.v. $\{(x, y)\}$

Let's first consider the question: how to qualify that **how well a model $f : \mathcal{X} \rightarrow \mathcal{Y}$ approximates the data (distribution of the pair r.v. z)?**

Given (1) the (empirical or population) distribution of a two-dimension pair r.v. $z = (x, y)$, (2) a loss function $\ell : \mathcal{F} \times \mathcal{Z} \rightarrow \mathbb{R}_{\geq 0}$, $\ell(f, z) := \ell(f(x), y)$

- ▶ empirical risk

$$\hat{\mathcal{R}}(f) = \frac{1}{n} \sum_{i=1}^n \ell(f, z_i)$$

- ▶ population risk

$$\mathcal{R}(f) = \mathbb{E}_{z \sim \mu}[\ell(f, z)]$$

Examples of loss function

- ▶ Mean Squared Loss: $\ell(f, (x, y)) = (f(x) - y)^2$
- ▶ Cross Entropy: $\ell(f, (x, y)) = -y \log(f(x)) - (1 - y) \log(1 - f(x))$

Note that the population risk equals $\mathbb{E}[\hat{\mathcal{R}}(f)]$ if we consider z to be an $n * 2$ -dimension r.v.[?]

Measurements of a function estimation

How well f_1 approximates f_2

Next: how to qualify **how well f_1 approximates f_2** ?

Moreover given (3) another model $f_2 : \mathcal{X} \rightarrow \mathcal{Y}$

- ▶ difference of population risks

$$\mathcal{R}(f_1) - \mathcal{R}(f_2) = \mathbb{E}_{z \sim \mu}[\ell(f_1, z) - \ell(f_2, z)]$$

which can be decomposed by

$$\begin{aligned}\mathcal{R}(f_1) - \mathcal{R}(f_2) &= [\mathcal{R}(f_1) - \hat{\mathcal{R}}(f_1)] + [\hat{\mathcal{R}}(f_1) - \hat{\mathcal{R}}(f_2)] + [\hat{\mathcal{R}}(f_2) - \mathcal{R}(f_2)] \\ &= \text{gen-gap}(f_1) + [\hat{\mathcal{R}}(f_1) - \hat{\mathcal{R}}(f_2)] - \text{gen-gap}(f_2)\end{aligned}$$

where the generalization gap is

$$\text{gen-gap}(f) = \mathcal{R}(f) - \hat{\mathcal{R}}(f) = \mathbb{E}_z[\ell(f, z)] - \frac{1}{n} \sum_{i=1}^n \ell(f, z_i)$$

Excess Risk

How well f approximates f^*

Suppose now we only consider function that is in a function class $\mathcal{F}$ (why?) and consider two special function, let f_1 and f_2 be

- ▶ $\hat{f}$ by empirical risk minimization $\hat{f} \in \arg \min_{f \in \mathcal{F}} \hat{\mathcal{R}}(f)$ So we have

$$\hat{\mathcal{R}}(\hat{f}) - \hat{\mathcal{R}}(f^*) = \frac{1}{n} \sum_{i=1}^n \ell(\hat{f}(x_i), y_i) - \frac{1}{n} \sum_{i=1}^n \ell(f^*(x_i), y_i) \leq 0$$

- ▶ f^* target model

The thing of interest in statistical learning is to analyze the behavior of the difference of population risks, which is called the **Excess Risk**.

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Uniform bound of gen gap

The Excess Risk can be decomposed by

$$\begin{aligned}\mathcal{R}(\hat{f}) - \mathcal{R}(f^*) &= \text{gen-gap}(\hat{f}) + [\hat{\mathcal{R}}(\hat{f}) - \hat{\mathcal{R}}(f^*)] - \text{gen-gap}(f^*) \\ &\leq \text{gen-gap}(\hat{f}) - \text{gen-gap}(f^*) \\ &\leq 2 \sup_{f \in \mathcal{F}} |\hat{\mathcal{R}}(f) - \mathcal{R}(f)|\end{aligned}$$

The question boils down to

→ the behavior of the uniform bound of gen-gap

$$\sup_{f \in \mathcal{F}} |\hat{\mathcal{R}}(f) - \mathcal{R}(f)| = \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \ell(f, z_i) - \mathbb{E}_{z \sim \mu} [\ell(f, z)] \right|$$

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Expected result:

A bound on $\sup_{f \in \mathcal{F}} \left| \hat{\mathcal{R}}(f) - \mathcal{R}(f) \right|$ in terms of $n, \mathcal{F}$, conditional on μ, ℓ .

Road Map:

1. Concentration: Consider $\sup_{f \in \mathcal{F}} \left| \hat{\mathcal{R}}(f) - \mathcal{R}(f) \right| = g(Z_1, \dots, Z_n)$, then by concentration inequalities, we can analyze its fluctuations.
[if the function is fluent enough]
2. Suprema: Define random process $\mathbb{X}_f := \frac{1}{n} \sum_{i=1}^n \ell(f, Z_i) - \mathbb{E}[\ell(f, Z)]$, then by theory of suprema, we can analyze its expectation through the properties of the function class.

Rademacher complexity and Gaussian complexity

Rademacher variable:

The Rademacher complexity of a set $A \subseteq \mathbb{R}^n$

$$\text{Rad}(A) = \mathbb{E}_{\varepsilon} \left(\sup_{\alpha \in A} |\langle \alpha, \varepsilon \rangle| \right)$$

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Case: binary classification

- ▶ data: $\{(x_i, y_i)\}_{i=1}^n \sim \mu$
- ▶ population: μ over $\mathcal{Z} = \mathcal{X} \times \{0, 1\}$
- ▶ function class: $\mathcal{F} \subseteq \{f : \mathcal{X} \rightarrow \{0, 1\}\}$
- ▶ loss function: for $z = (x, y)$, $\ell(f, z) = \mathbf{1}\{f(x) \neq y\} \in \{0, 1\}$

Since cross entropy loss is bounded, we have

$$\sup_{f \in \mathcal{F}} |\hat{\mathcal{R}}(f) - \mathcal{R}(f)| = g(Z_1, \dots, Z_n), \quad \|Dig\|_\infty \leq \frac{2}{n},$$

Then by McDiarmid, for any $t > 0$,

$$\mathbb{P}\{g(Z_1, \dots, Z_n) - \mathbb{E}[g(Z_1, \dots, Z_n)] \geq t\} \leq \exp(-nt^2/2).$$

Equivalently, $\forall \delta \in (0, 1)$,

$$\mathbb{P}(\sup_{f \in \mathcal{F}} |\hat{\mathcal{R}}(f) - \mathcal{R}(f)| \leq \mathbb{E} \left[\sup_{f \in \mathcal{F}} |\hat{\mathcal{R}}(f) - \mathcal{R}(f)| \right] + \sqrt{\frac{2 \log(1/\delta)}{n}}) \geq 1 - \delta$$

Case: binary classification

[Symmetrization of Empirical Processes]

$$\begin{aligned}\mathbb{E} \left[\sup_{f \in \mathcal{F}} \left| \hat{\mathcal{R}}(f) - \mathcal{R}(f) \right| \right] &\leq \frac{2}{n} \mathbb{E}_{Z, \varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \sum_{i=1}^n \varepsilon_i \ell(f, Z) \right| \right] \\ &= \frac{2}{n} \mathbb{E}_{Z, \varepsilon} \left(\sup_{\alpha \in S(Z)} |\langle \alpha, \varepsilon \rangle| \right) \\ &\leq \frac{2}{n} \mathbb{E}_Z [\text{Rad}(S(Z))]\end{aligned}$$

where

$$S(Z) = \left\{ (\ell(f, Z_1), \dots, \ell(f, Z_n)) : f \in \mathcal{F} \right\} \subseteq [0, 1]^n$$

Case: binary classification

Case: regression

OLS with fixed design

References

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